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Quivers, cones and polytopes

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Dedicated to Idun Reiten on the occasion of her 60th birthday

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Abstract

Let Q be a quiver without oriented cycles. We consider the polytope of flows $\Delta(\theta)$ in Q with input θ . These polytopes are closely related to the combinatorial structure of the quiver, in particular, to its spanning subtrees. Furthermore, we consider a system of cones which turns out to be a fan and can be seen as a base for the family of all flow polytopes $\Delta(\theta)$ for the various inputs θ . Finally, we present several examples.

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1. Introduction

Let Q be a finite quiver without oriented cycles. A weight of Q is a real function on the set of vertices of Q , whose values add up to zero (for precise definitions we refer to Section 2). We denote the space of all weights by $\mathbb{H}_{\mathbb{R}}$. To the quiver Q we associate various combinatorial objects:

- for any weight θ a polytope $\Delta(\theta)$,
- for any subquiver P of Q a cone C_P in $\mathbb{H}_{\mathbb{R}}$,
- for any weight θ a cone $\bar{C}(\theta)$ in $\mathbb{H}_{\mathbb{R}}$, and
- a set of hyperplanes in $\mathbb{H}_{\mathbb{R}}$ called walls.

The principal aim of this note is to investigate these combinatorial objects and to relate them to the other ones. The polytopes $\Delta(\theta)$ were already introduced in

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[1]. The cones and the walls above control the combinatorial change of the polytope $\Delta(\theta)$ if we vary the weight θ . The cones C_P control the change of $\Delta(\theta)$ locally (a face of $\Delta(\theta)$ corresponds to a subquiver P of Q precisely when the weight θ is in C_P , see Theorem 3.2), whereas the cones $\overline{C}(\theta)$ control the change of $\Delta(\theta)$ globally (the faces of $\Delta(\theta)$ correspond to subquivers $P^1, \dots, P^r$ precisely when θ is in the intersection of the cones C_{P^i} for the various i). Further, it turns out that each facet of the cone C_P and the cone $\overline{C}(\theta)$, respectively, is contained in some wall. Conversely, the set of walls is the minimal set of hyperplanes with this property. Finally the walls control the sign of the solutions of the flow equation (1) along trees (these solutions are called virtual vertices, see Proposition 3.3 and its proof).

Using toric geometry (see e.g. [4,6,8]) the polytopes $\Delta(\theta)$ have an algebraic-geometric interpretation (see [1,5], and [7] for an interpretation as moduli space). The change of the polytope $\Delta(\theta)$ corresponds to a flip (see e.g. [9,10] for a general approach and [5] for results on moduli spaces of thin sincere representations). Certain polytopes, we shall consider in a forthcoming paper, are related to torus actions on flag varieties and to toric varieties which are degenerations of flag varieties (see also [3]). Furthermore, we mention reflexive polytopes (see Section 5, final remark). Reflexive polytopes are used in Batyrev's construction of mirrors in toric varieties [2]. For examples we refer to Section 6 and Example 1, Section 3. We note that $\Delta(\theta)$ is reflexive for the canonical weight [1, Proposition 2.7]. A different construction of a toric variety can be obtained from Theorem 5.1. Since the cones $\overline{C}(\theta)$ form a fan, there exists a toric variety X associated to this fan (see the standard references on toric geometry [4,6,8]). However, so far there exist no further results concerning this toric variety X .

This paper is organized as follows. In Section 2 we start with our basic notation. In Section 3 we study the polytopes $\Delta(\theta)$ for a quiver Q without oriented cycles. In particular, we relate the faces of $\Delta(\theta)$ to the spanning subquivers P and we relate the combinatorial properties of the polytope $\Delta(\theta)$ to the combinatorics of the subquivers of Q . Further we study the variation of the combinatorial structure when the weight θ varies. In Section 4 we investigate the cones C_P and relate them to the walls. In particular, we determine the minimal set of primitive vectors, so that each cone C_P is generated by a subset of these vectors. Further, in Section 5 we construct a system of cones $\overline{C}(\theta)$. Our main result in this section, Theorem 5.1, states that these cones form a so-called fan. Finally, in Section 6 we consider some examples to illustrate the constructions in the previous sections.

2. Basic notation

2.1. Quivers and polytopes

Let Q be a finite connected quiver without oriented cycles. We denote by Q_0 the vertices of Q (usually identified with the natural numbers $1, \dots, \sharp Q_0$) and by Q_1

the arrows. For an arrow α we denote its starting vertex with $s(\alpha)$ and its terminal vertex with $t(\alpha)$. Further we associate to any quiver certain affine spaces: the vector space $\mathbb{H}_{\mathbb{R}}$ of weights in Q , the vector space of flows $\mathbb{R}^{Q_1}$ and the affine space $M(\theta)_{\mathbb{R}}$ of all flows with a certain fixed weight θ . All these spaces have an integral counter part and contain certain cones defined by the quiver Q . The vector space $\mathbb{H}_{\mathbb{R}}$ consists of all elements θ in $\mathbb{R}^{Q_0}$ which satisfy one relation

$$\sum_{q \in Q_0} \theta(q) = 0.$$

There exists a natural map $I_Q : \mathbb{R}^{Q_1} \longrightarrow \mathbb{R}^{Q_0}$ sending the flow ε to its input θ , defined by

$$\theta(q) := \sum_{\alpha \mid s(\alpha)=q} \varepsilon(\alpha) - \sum_{\alpha \mid t(\alpha)=q} \varepsilon(\alpha). \quad (1)$$

The image of this map is contained in $\mathbb{H}_{\mathbb{R}}$ (just add up all these equations) and we obtain an exact sequence

$$0 \longrightarrow M(0)_{\mathbb{R}} \longrightarrow \mathbb{R}^{Q_1} \xrightarrow{I_Q} \mathbb{H}_{\mathbb{R}} \longrightarrow 0.$$

Further we define $M(\theta)_{\mathbb{R}} := I_Q^{-1}(\theta)$. Note that I_Q is an integral map. Thus, the vector space of flows contains the lattice $\mathbb{Z}^{Q_1}$, the space of weights contains the lattice $\mathbb{H} := \mathbb{H}_{\mathbb{R}} \cap \mathbb{Z}^{Q_0}$, and the map I_Q restricts to a lattice homomorphism. Moreover, for θ integral (that is for θ in $\mathbb{H}$), $M(\theta) := M(\theta)_{\mathbb{R}} \cap \mathbb{Z}^{Q_1}$ is a lattice in $M(\theta)_{\mathbb{R}}$. Note that $\mathbb{R}^{Q_1}$ contains the cone $\mathbb{R}_{\geq 0}^{Q_1}$ of so-called regular flows. We are mainly interested in the intersection

$$\Delta(\theta) := M(\theta)_{\mathbb{R}} \cap \mathbb{R}_{\geq 0}^{Q_1}$$

of the regular flows with the affine space $M(\theta)_{\mathbb{R}}$. For a quiver without oriented cycles the space $\Delta(\theta)$ is bounded and an intersection of certain half spaces. Consequently, it is a polytope. Moreover, for θ integral, $\Delta(\theta)$ is generated by lattice points (Lemma 3.1).

2.2. The set of walls

The combinatorial structure of the polytope $\Delta(\theta)$ is closely related to the combinatorial structure of the subquivers of Q . We introduce a set of certain hyperplanes in $\mathbb{H}_{\mathbb{R}}$ which turn out to control the combinatorial change of $\Delta(\theta)$ if we vary the parameter θ : let $Q_0 = Q_0^+ \cup Q_0^-$ be a decomposition of the vertices of Q into two disjoint non-empty subsets. Denote by Q^+ and Q^- the full subquivers of Q with vertices in Q_0^+ , respectively, in Q_0^- . We further assume both Q^+ and Q^- are connected. For those subdivisions we define two half spaces and a hyperplane

$$W^+ := \left\{ \theta \in \mathbb{H}_{\mathbb{R}} \mid \sum_{q \in Q_0^+} \theta(q) \geq 0 \right\};$$

$$W^- := \left\{ \theta \in \mathbb{H}_{\mathbb{R}} \mid \sum_{q \in Q_0^+} \theta(q) \leq 0 \right\}; \quad (2)$$

$$W := W^+ \cap W^-.$$

The hyperplane W is called a *wall*, the half space W^+ is called the *positive half space*, and the half space W^- is called the *negative half space* of the wall W . Note that we can define these spaces for any subdivision of the vertices of Q , that is without the connecting assumption above, however, only the walls will play a particular role in our considerations (see Theorem 3.2 and Proposition 3.3). For any wall W we define the *type* (t^+, t^-) of the wall, where t^+ is the number of arrows starting in Q^+ and ending in Q^- and t^- is the number of arrows starting in Q^- and ending in Q^+ . Such a wall we also call (t^+, t^-) -wall. Further, we assume without loss of generality $t^+ \geq t^-$. Note that this convention provides any wall of non-symmetric type (that is $t^+ \neq t^-$) with a natural orientation. A wall of type $(t^+, 0)$ is also called an *outer wall*, a wall of type (t^+, t^-) for $t^- > 0$ is called an *inner wall*.

2.3. Quivers and cones

Further we are interested in certain cones in $\mathbb{H}_{\mathbb{R}}$. For let $P \subset Q$ be a subquiver of Q which has the same vertices as Q and is also assumed to be connected. Thus, P contains a spanning tree. We consider the cone D_P of all regular flows along P :

$$D_P := \{ \varepsilon \text{ in } \mathbb{R}^{Q_1} \mid \varepsilon(\alpha) = 0 \text{ for all } \alpha \notin P_1, \quad \varepsilon(\alpha) \geq 0 \text{ for all } \alpha \in Q_1 \}.$$

The dimension of this cone equals the number of arrows in P . We consider its image C_P under the projection I_Q to the weight space $\mathbb{H}_{\mathbb{R}}$. It turns out that these cones C_P are all of maximal dimension, that is its dimension equals the dimension of $\mathbb{H}_{\mathbb{R}}$, which is the number of vertices minus one. The walls W and the cones C_P for the various subquivers are closely related: for quivers without any $(1,0)$ -wall and any $(1,1)$ -wall the vector spaces generated by the facets of the cones are precisely the walls (Lemma 4.2). Further, it is easy to see that

$$C_P = \bigcup_{T \subseteq P} C_T,$$

where the union runs over the connected subtrees of P with $T_0 = P_0 = Q_0$ (Lemma 4.1). For any weight θ we define

$$C(\theta) := \bigcap_{\theta \in C_P} C_P$$

and

$$\overline{C}(\theta) := C(\theta) \cap \bigcap_{W \text{ an outer wall} \mid \theta \in W} W.$$

It seems to be natural to ask similar questions in more general situations: we can replace the linear map I_Q by any surjective integral map. For those general maps one can define, in an analogous way, a set $\Delta(\theta)$, walls and cones. However there appear several natural problems. First, the set $\Delta(\theta)$ may in general not be bounded. Even if it is bounded, the vertices are in general not lattice points. In particular, the polytope for the canonical weight is in general not reflexive, since it is not a lattice polytope. Furthermore, the walls and the cones are much harder to describe.

2.4. Notation

We assume a quiver is always finite, connected and without oriented cycles. Moreover, we assume that the quiver Q is *reduced*, that is it admits no $(1,0)$ -wall and no $(1,1)$ -wall. A subquiver of a quiver Q is always assumed to have the same vertices as Q . A tree is a connected and simply connected quiver. Further we define the dimension $\dim Q$ of a quiver Q to be the dimension of its first singular homology group, which equals the number of arrows subtracted by the number of vertices plus one.

Polytopes and cones are considered to be subsets of some real vector space, where we assume that 0 is the apex of each cone. We assume this vector space always contains a lattice. A point in the lattice is called a lattice point. For any polytope Δ , a *vertex* is a point which admits a half space V^+ containing x in its boundary so that the intersection of Δ with V^+ is precisely the point x . Similar for a cone C : a vertex x in C is an *extremal point* if there exists a halfspace V^+ with x (and 0) in its boundary, so that V^+ intersected with C is precisely the ray starting in zero through x . A *face* of a polytope (or a cone) is just the intersection of the polytope (or the cone, respectively) with some half space V^+ , whose interior does not intersect the polytope (the cone, respectively). Such a half space is called a *supporting half space* and its boundary is called a *supporting hyperplane*. A facet is a face of codimension one. Any polytope is the convex hull of the finite number of vertices. Also, any polytope is the intersection of all half spaces containing the polytope. Similarly, any polyhedral cone is generated as a cone by a finite number of its extremal points. It is also the intersection of all the half spaces containing it. We assume, in particular, that 0 is a face of each cone. A cone is called *simplicial* if there exists a set of extremal points, which generate the cone and are linearly independent over $\mathbb{R}$. It is called *smooth*, if in addition these extremal points can be chosen as lattice points and form a part of a $\mathbb{Z}$ -basis of the lattice.

If we say two polytopes Δ and Δ' are combinatorially equivalent, then we mean that their posets of faces are isomorphic as posets. Note that the faces of a polytope are ordered by the obvious incidence relation. For quiver polytopes this poset is provided with a map of the maximal elements in the face poset (consisting of facets) to the arrows of the quiver (see Theorem 3.2, and take the facet corresponding to the subquiver $Q \setminus \{\alpha\}$). The combinatorial equivalence, we consider in this note, does also preserve this map.

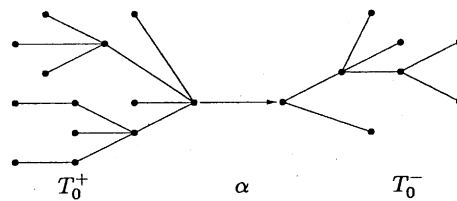


Fig. 1.

3. Quivers and polytopes

Note that the weight θ can be seen as the input in the “network” Q with flow ε . Then the flow and the input are related via the *flow equation* (1). Assume for a moment the quiver is just a tree T , then the flow equation for a fixed input θ admits a unique flow $\varepsilon^T(\theta)$. Explicitly we obtain

$$\varepsilon^T(\theta)(\alpha) = \sum_{q \in T_0^+(\alpha)} \theta(q) = - \sum_{q \in T_0^-(\alpha)} \theta(q), \quad (3)$$

where $T_0^+(\alpha)$ consists of all points connected to $s(\alpha)$ in $T \setminus \{\alpha\}$ and $T_0^-(\alpha)$ consists of the complement of $T_0^+(\alpha)$ in T_0 (see Fig. 1). In other words, the flow in an edge of the tree T equals the sum of the inputs in one of the connected components and it also equals the sum of the outputs in the other component. In particular, the tree T admits a regular flow precisely when for each arrow α the sum of inputs in the component connected with the starting point of α is nonnegative. We illustrate this fact in Fig. 1.

The latter statements in Lemmas 3.1 and 4.1 are well-known results in discrete optimization. Both results also follow easily from the fact, that each minor of the matrix I_Q is -1 , 0 , or 1 .

Lemma 3.1. *The set $\Delta(\theta)$ is a polytope. Each subtree T of Q , which admits a regular flow along T , defines a vertex $\varepsilon^T(\theta)$ in $\Delta(\theta)$. Conversely, each vertex of $\Delta(\theta)$ is of this form.*

Proof. The set $\Delta(\theta)$ is bounded, since each coordinate of a regular flow is bounded by the sum of all positive inputs (here we use that Q has no oriented cycles), and an intersection of half spaces. Thus, it is a polytope. Note that any point ε on the boundary of the polytope $\Delta(\theta)$ satisfies an equation $\varepsilon_\alpha = 0$ for some arrow α . Obviously, a regular flow $\varepsilon^T(\theta)$ satisfies as many of those equations as possible. Thus, any regular flow $\varepsilon^T(\theta)$ in the polytope $\Delta(\theta)$ must lie on the boundary and is extremal. Conversely, any vertex ε in $\Delta(\theta)$ is defined by a maximal family of equations $\varepsilon_\alpha = 0$. Thus, the support of ε is contained in some tree T and we obtain $\varepsilon = \varepsilon^T(\theta)$. $\square$

Note, that in general there is not a bijection between the set of subtrees admitting a regular flow with the set of vertices: different trees can admit the same flow in Q (see, e.g., Section 6, Examples 1 and 2), since the support of $\varepsilon^T(\theta)$ may coincide with the support of $\varepsilon^{T'}(\theta)$ for some $T \neq T'$. However, for a generic weight θ we obtain a bijection. We want to make the genericity condition more precise: a weight θ is *generic* if it is an element of C_Q and does not lie on any of the facets of any of the cones C_T for the various subtrees T . Note that our definition is different from the one used in [5] since any genericity condition depends on the actual problem one is interested in (see also Proposition 3.3 and the note above it for a further explanation): for the classification of combinatorial classes of polytopes we need the genericity condition defined above.

Theorem 3.2. *The polytope $\Delta(\theta)$ is of dimension $\sharp Q_1 - \sharp Q_0 + 1$ for any θ in the interior of C_Q and it is nonempty precisely if θ is in C_Q . If θ is in generic position, then the vertices of $\Delta(\theta)$ are in bijection with those subtrees of Q admitting a regular flow and the faces F of $\Delta(\theta)$ are in bijection with the subquivers P of Q admitting a regular flow. Moreover, $\dim F = \dim P$ for the face F defined by P .*

Proof. Remember the definition of C_Q , it consists of all weights θ , so that there exists a regular flow with input θ . Consequently, $\Delta(\theta)$ is empty precisely when θ is not in C_Q . Counting dimensions of D_Q and C_Q show

$$\Delta(\theta) = \dim D_Q - \dim C_Q = \sharp Q_1 - (\sharp Q_0 - 1).$$

Here we use the fact that D_Q projects surjectively onto C_Q , both are convex cones, and the map is linear. In particular, the fibre of this map is convex as well and the dimension of the fibre is the same for all elements θ in the interior of C_Q . This proves the claim on the dimension of $\Delta(\theta)$.

Assume $\varepsilon^T(\theta)$ and $\varepsilon^{T'}(\theta)$ coincide as flows in Q for some trees $T \neq T'$. Then $\varepsilon^T(\theta) = \varepsilon^{T \cap T'}(\theta) = \varepsilon^{T'}(\theta)$. That is, the flow equation admits a solution even on the intersection of the two trees T and T' . The boundary of C_T consists of all those weights whose unique flow $\varepsilon^T(\theta)$ vanishes on some arrow α in T . Consequently, since $\varepsilon^T(\theta) = \varepsilon^{T'}(\theta)$ as above, the weight θ is in the boundary of C_T and also in the boundary of $C_{T'}$. Conversely, any flow $\varepsilon^T(\theta)$ for a weight θ in the boundary of some cone C_T coincides with some flow $\varepsilon^{T'}(\theta)$ for $T \setminus \{\alpha\} \subset T'$. Such a tree $T' \neq T$ exists for any reduced quiver Q , in fact it is sufficient that Q admits no $(1, 0)$ -wall.

To show the last claim we note that any face F is the convex hull of its vertices $\varepsilon_1, \dots, \varepsilon_r$. Thus $\varepsilon_i = \varepsilon^{T^i}(\theta)$ for certain trees T^i . These trees are pairwise different and uniquely determined, since θ is generic. Consider $P = \bigcup_{i=1}^r T^i \subset Q$, the subquiver consisting of all arrows of the various T^i . Then

$$F = \{\varepsilon \in \Delta \mid \varepsilon(\alpha) = 0 \text{ for all } \alpha \notin P\}.$$

This way, any face of $\Delta(\theta)$ defines a unique subquiver P which obviously admits a regular flow. Conversely, let P be any subquiver which admits a regular flow. Let T^i be the subtrees of P admitting a regular flow ε^i . Then these flows are pairwise

different, since θ is generic. Moreover, θ is also generic for P , thus $\dim P = \dim F$ and the proof is finished. $\square$

A similar result can be proven for “virtual vertices”, these are the flows along trees which are not regular. For those virtual vertices we have to replace the faces of the cones C_T by the walls (see the proposition below).

Proposition 3.3. *The elements $\varepsilon^T(\theta)$ are all different precisely when θ does not lie on any of the walls W .*

Proof. Assume θ does not lie on any wall and assume $\varepsilon^T(\theta) = \varepsilon^{T'}(\theta)$ for some trees $T \neq T'$. Then $\varepsilon^T(\theta)(\alpha) = 0$ for α in T but not in T' . Consequently, for the subdivision of T in Fig. 1 we obtain

$$\sum_{q \in T_0^+} \theta(q) = 0.$$

Obviously T^+ and T^- are both connected and

$$W := \left\{ \eta \in \mathbb{H}_{\mathbb{R}} \mid \sum_{q \in T_0^+} \eta(q) = 0 \right\}$$

is a wall containing θ , a contradiction. Conversely, let θ be an element of some wall W , where W is defined by the subdivision $Q_0 = Q_0^+ \cup Q_0^-$ (see the definition in Section 1). Let T^+ be a subtree of Q^+ and T^- a subtree of Q^- . Since the quiver Q is reduced, there exists no (1,0)-wall for Q . Thus, there exist two arrows α and β both have its starting and terminal vertex in different subsets. e.g., for a (2,0)-wall there exist arrows α and β with $s(\alpha)$ and $s(\beta)$ in Q_0^+ , and $t(\alpha)$ and $t(\beta)$ in Q_0^- . Let T be $T^+ \cup T^- \cup \{\alpha\}$ and let T' be $T^+ \cup T^- \cup \{\beta\}$. Both T and T' are trees and $\varepsilon^T(\theta) = \varepsilon^{T'}(\theta)$, since both flows vanish on α and β . $\square$

Note that the proposition is also true for a quiver with a (1,1)-wall.

Example 1. We consider the hexagon defined by the following quiver (left-hand side) with weight indicated on the vertices. On the quiver acts the product of two symmetric groups $S_3 \times S_2$ via permutation of the sources and the sinks, respectively. So this group also acts on the 12 trees of the quiver and there exist precisely two orbits (Fig. 2, right-hand side), each consists of six trees. Since the weight is invariant under the action, the flows along the trees are just permuted. In each of the two classes of trees we choose one representative with its unique flow. It turns out that one class admits a regular flow (right-hand side, below), so we have six vertices (indicated by $\bullet$), and the other class does not admit a regular flow (right-hand side, above), so we have six additional virtual vertices (indicated by $\circ$).

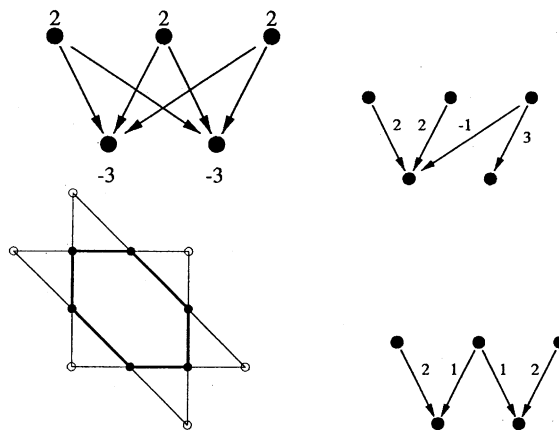


Fig. 2.

4. The cones C_P

Our main interest in this section concerns the cones C_P for the various subquivers P of the quiver Q . Recall that P is always assumed to be connected and with the same vertices as Q . We start this section with some results on the structure of the cones C_T for trees T . It is obvious that the cones D_P in $\mathbb{R}^{Q_1}$ are simplicial, that is, generated by a part of an $\mathbb{R}$ -basis. Even better, they are smooth, that is, the cone is generated as a cone by a part of the $\mathbb{Z}$ -basis of the integral lattice $\mathbb{Z}^{Q_1}$. The situation for the cones C_P is more complicated. However, a similar result holds for trees, whereas for arbitrary subquivers P it may not hold (see Lemma 4.4 and Example 4 with Fig. 9).

Lemma 4.1. *Let T be a (connected) subtree of Q (with $T_0 = Q_0$). Then the cone C_T is smooth. Moreover, $\dim C_T = \dim \mathbb{H}_{\mathbb{R}}$.*

Proof. For the proof, we introduce the integral elementary flows

$$\phi_{\alpha}(\beta) = \begin{cases} 1, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

The image of ϕ_{α} under I_Q is the elementary weight $\omega_{t(\alpha)}^{s(\alpha)}$, defined by

$$\omega_q^p(i) := \begin{cases} 1, & i = p, \\ -1, & i = q, \\ 0 & \text{else} \end{cases}$$

for any vertices p, q , and i in Q . Obviously, any regular flow admits a unique decomposition into elementary flows with nonnegative coefficients. Consequently, any weight of a regular flow also admits a decomposition into elementary weights with positive coefficients, however this decomposition in general is not unique. Similar

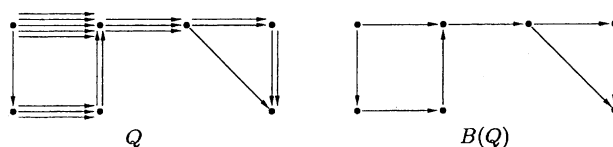


Fig. 3.

arguments show the existence of a decomposition with integral coefficients for any integral flow and any integral weight, since any integral regular flow admits a unique decomposition into elementary flows with nonnegative integral coefficients. Thus, for any subquiver P , we already know, that any integral weight in C_P is in the $\mathbb{Z}_{\geq 0}$ -span of the elementary weights $\omega_{t(\alpha)}^{s(\alpha)}$, where α runs through the arrows of P . If this subquiver is a tree T , then the elements $\omega_{t(\alpha)}^{s(\alpha)}$ for α in T_1 are linearly independent. In fact, the matrix consisting of the vectors of $\omega_{t(\alpha)}^{s(\alpha)}$ is of rank $\sharp Q_0 - 1$. Viewed as a matrix of an endomorphism of $\mathbb{H}_{\mathbb{R}}$ its determinant is 1 or -1 . $\square$

Lemma 4.2. *Any facet of a cone C_T is the intersection of some wall W with C_T . This wall is determined by some subdivision of Q with $T_0^+ = Q_0^+$ and $T_0^- = Q_0^-$ defined by an arrow α in T (see Fig. 1).*

Proof. The number of facets of C_T is $\sharp Q_0 - 1 = \sharp T_1$. Clearly $C_T \cap W$ is a facet of C_T for any wall as above and for different walls we get different facets: the intersection of two walls is already of codimension 2. Consequently, we already got all facets of C_T in this way. $\square$

Note that the intersection of C_T with any wall may not be a face of C_T , see Section 6, Example 2.

For any quiver Q we define a quiver $B(Q)$ as the quiver obtained from Q by deleting multiple arrows. Thus, $B(Q)_0 = Q_0$ and there is precisely one arrow α in $B(Q)$ starting in p and ending in q if there exists an arrow in Q starting in p and ending in q . We illustrate the construction in Fig. 3.

For the rest of the paper we identify the weight spaces of Q and $B(Q)$.

Corollary 4.3. *Let Q and Q' be quivers with $B(Q) \simeq B(Q')$. Then the cones C_P in $\mathbb{H}$ only depend on the image of P in $B(Q)$. In particular, the system of cones C_P , the walls W and the generacity condition on θ depend only on the quiver $B(Q)$ of the quiver Q .*

Proof. In the proof of Lemma 4.1 we introduced the elementary weights as images of the elementary flows. Obviously, the elementary weights only depend on the $B(Q)$ -class of Q . $\square$

Remark. Note that for a reduced quiver Q , the quiver $B(Q)$ is not necessarily reduced. So the cones for Q and $B(Q)$ may differ. As an example we consider the Kronecker quiver R_t with two vertices and t arrows starting in one of the vertices and ending in the other one. For $t \geq 2$ the quiver R_t is reduced, whereas $B(Q) = R_1$ is not reduced.

Lemma 4.4. *For any quiver Q the cone C_Q is the union of the cones C_T , where T runs through the subtrees of Q .*

Proof. Let ε be a regular flow, that is ε is in D_P . Such a flow admits a decomposition

$$\varepsilon = \sum_{\alpha \in Q_1} a_\alpha \phi_\alpha$$

into elementary flows with nonnegative coefficients a_α . For any α in P , there exists a tree $T \subset P$ with α in T . Thus $D_P = \bigcup_T D_T$ and, consequently, after projection under I_Q we obtain $C_P = \bigcup_T C_T$, where T runs through all subtrees of P . $\square$

Note that it is sufficient to take the union over some representatives of the trees in $B(Q)$ which contain all the arrows in $B(Q)$.

Lemma 4.5. *Any facet of the cone C_Q is of the form $C_Q \cap W$ for some outer wall W . Moreover, each face of C_Q is the intersection of C_Q with some outer walls. Conversely, any intersection of C_Q with any set of outer walls is a face of C_Q .*

Proof. By definition of an outer wall W the polytope $\Delta(\theta)$ is empty for θ in the interior of W^- . Thus W is a supporting hyperplane for C_Q . Also $C_Q = \bigcap W^+$, where the intersection runs over all positive half spaces of outer walls W . Consequently, each facet is of the form $C_Q \cap W$ and, finally, each face is an intersection of facets. $\square$

If we apply the result above to a subquiver P , then outer walls for P are certainly walls for Q . However, for any wall W the intersection $C_P \cap W$ is not necessarily a face of C_P . This only holds for walls W , which are outer walls for P . Thus we already have proven:

Corollary 4.6. *For any subquiver P of Q each facet of C_P is of the form $C_P \cap W$ for some wall W of Q . Moreover, this wall W is an outer wall for P .*

To investigate the structure of the cones C_P for any subquiver P of Q we need some more notation. We say an arrow α in P is *primitive* if there does not exist a path $w = \beta_1 \cdots \beta_r$ in P with $r \geq 2$, $s(w) = s(\alpha)$, and $t(w) = t(\alpha)$. In other words there exists no detour in P for a flow through α . Note that for trees each arrow is primitive.

Proposition 4.7. *The cone C_P is minimally generated as a cone by the elementary weights $\omega_{t(\alpha)}^{s(\alpha)}$, where α runs through the set of primitive arrows in P .*

Proof. Let α be not primitive. Then $s(\alpha) = s(w)$ and $t(\alpha) = t(w)$ for some path $w = \beta_1 \cdots \beta_r$, where $r \geq 2$. Thus

$$\omega_{t(\alpha)}^{s(\alpha)} = \sum_{i=1}^r \omega_{t(\beta_i)}^{s(\beta_i)}$$

and $\omega_{t(\alpha)}^{s(\alpha)}$ is not needed to generate the cone C_P . Conversely, if α is primitive and $\omega_{t(\alpha)}^{s(\alpha)}$ is the corresponding elementary weight then we show the weight $\omega_{t(\alpha)}^{s(\alpha)}$ is extremal in C_P . Assume the converse. Then there exist (even primitive) arrows $\beta_1, \dots, \beta_r$ so that

$$\omega_{t(\alpha)}^{s(\alpha)} = \sum_{i=1}^r a_i \omega_{t(\beta_i)}^{s(\beta_i)}$$

with $a_i > 0$. If we compare the coefficients with respect to any vertex q in the equation above, we obtain (after reordering the arrows β_i) $s(\alpha) = s(\beta_1)$, $t(\beta_1) = s(\beta_2)$, $\dots$, $t(\beta_{i-1}) = s(\beta_i)$, and finally $t(\alpha) = t(\beta_i)$. Thus, α is not primitive, a contradiction. $\square$

5. The cones $\overline{C}(\theta)$

Further, we are interested in the system of cones generated by all the cones C_P for the various subquivers P of Q . For we define for any weight θ the cone $C(\theta)$ which is the intersection of all cones C_P containing θ . According to Lemma 4.4 this cone is

$$C(\theta) := \bigcap_{T \mid T \in \mathcal{T}(\theta)} C_T,$$

$$\text{where } \mathcal{T}(\theta) := \{T \mid \varepsilon^T(\theta)(\alpha) \geq 0\} = \{T \mid \theta \in C_T\}.$$

It turns out that these cones $C(\theta)$ are no longer generated by elementary weights of arrows (see Example 2, Section 6, and Figs. 9 and 10).

To state our main result in this section we have to recall the notion of a fan in $\mathbb{H}_{\mathbb{R}}$. Recall that a cone C (with apex in 0) is called *convex*, if for any x, y in C , also $\lambda x + (1 - \lambda)y$ for any λ in $[0, 1]$ is in C . It is called *polyhedral*, if it is generated by a finite number of elements as a convex cone. Moreover, it is *strongly convex*, if C does not contain a line (one-dimensional affine subspace) and, finally, it is called *rational* if C is generated by lattice points. A *fan* consists of a finite set of strongly convex, rational, polyhedral cones $\Sigma = \{\sigma_i \mid i \in I\}$ which satisfy the following two conditions (see [4], [6], or [8]):

- (1) the face of any cone in Σ is also a cone in Σ , and
- (2) the intersection of two cones in Σ is a face of both.

The cones $C(\theta)$ introduced above only satisfy the second condition of a fan, since the face of a cone $C(\theta)$ in general is not the intersection of cones of the form C_T . So we have to work with the cones $\overline{C}(\theta)$ instead.

Theorem 5.1. *The system of cones $\{\overline{C}(\theta)\}$ for θ in C_Q is a fan.*

To prove the theorem we proceed in several steps. First we prove some results concerning the intersection of the cones $C(\theta)$. For two weights θ and θ' , we say, they are *separated* by the wall W if θ is in the interior of one of the half spaces W^+ and θ' is in the interior of the other one W^- , or vice versa. Similarly, we say a wall W separates two cones $C(\theta)$ and $C(\theta')$ if W separates any weight η in the interior of $C(\theta)$ from any weight η' in the interior of $C(\theta')$.

Lemma 5.2. *If θ and θ' are generic. Then $\dim C(\theta) = \dim C(\theta') = \dim \mathbb{H}_{\mathbb{R}}$. Assume $C(\theta)$ and $C(\theta')$ are separated by precisely one wall W . Then*

$$C(\theta) \cap W = C(\theta') \cap W = C(\theta) \cap C(\theta').$$

Moreover, let η be a weight in the interior of $C(\theta) \cap C(\theta')$. Then $C(\eta) = C(\theta) \cap C(\theta')$.

Proof. The claim about the dimension can be seen as follows. Each cone C_T is of this dimension by Lemma 4.1. A generic weight is an interior point of each cone C_T . Consequently, θ is in the interior of any finite intersection of those cones, in particular, it is in the interior of $C(\theta)$.

For the proof, we compare the sets $\mathcal{T}(\theta)$, $\mathcal{T}(\theta')$ and $\mathcal{T}(\eta)$. Let $Q_0^+ \cup Q_0^-$ be the subdivision defining W , so

$$W = \left\{ \xi \in \mathbb{H} \mid \sum_{q \in Q_0^+} \xi(q) = 0 \right\}.$$

We further assume, to have a preferred orientation, $\sum_{q \in Q_0^+} \theta(q) > 0$. Thus, $\sum_{q \in Q_0^+} \theta'(q) < 0$. We define a subset $\mathcal{T}(\theta)^+$ of $\mathcal{T}(\theta)$ consisting of those trees T which contain precisely one arrow α connecting a vertex in Q_0^+ with a vertex in Q_0^- . Such an arrow must start in Q_0^+ and end in Q_0^- . Similarly, for $\mathcal{T}(\theta')$ we define a subset $\mathcal{T}(\theta')^-$ consisting of trees containing precisely one arrow α connecting a vertex in Q_0^+ with a vertex in Q_0^- . Such a vertex must start in Q_0^- and end in Q_0^+ . We claim

$$\mathcal{T}(\theta) \cap \mathcal{T}(\theta') = \mathcal{T}(\theta) \setminus \mathcal{T}(\theta)^+ = \mathcal{T}(\theta') \setminus \mathcal{T}(\theta')^-.$$

Just by definition and using Eq. (3) any tree in $\mathcal{T}(\theta)^+$ is not in $\mathcal{T}(\theta')$. Let T be a tree in $\mathcal{T}(\theta) \setminus \mathcal{T}(\theta)^+$ which is not in $\mathcal{T}(\theta')$. Then $\varepsilon^T(\theta')(\alpha) < 0$ for some arrow α

in T . The subdivision in Fig. 1 defines a wall different from W which separates θ and θ' . Thus this new wall separates also $C(\theta)$ from $C(\theta')$, a contradiction. Using the definition of $C(\theta)$ in terms of trees finishes the proof of the first claim.

To show the second claim, we show $\mathcal{T}(\eta) = \mathcal{T}(\theta) \cup \mathcal{T}(\theta')$. Obviously $\mathcal{T}(\theta) \cup \mathcal{T}(\theta') \subseteq \mathcal{T}(\eta)$. Let T be a tree in $\mathcal{T}(\eta)$ which is not in $\mathcal{T}(\theta) \cup \mathcal{T}(\theta')$. We claim $\varepsilon^T(\eta)(\alpha)$ must be zero for two different arrows α in T : assume $\varepsilon^T(\eta)(\alpha) = 0$ only for one arrow. Then $\varepsilon^T(\theta)(\alpha) > 0$ or $\varepsilon^T(\theta')(\alpha) > 0$, so T is in $\mathcal{T}(\theta) \cup \mathcal{T}(\theta')$, a contradiction. So we can assume $\varepsilon^T(\eta)(\alpha_1) = \varepsilon^T(\eta)(\alpha_2) = 0$ for two different arrows α_1 and α_2 in T . This is a contradiction, since then η is an element of two different walls defined by the subdivisions of Fig. 1 for α_1 and α_2 . This finishes the proof. $\square$

Lemma 5.3. *Each face of a cone $C(\theta)$ is of the form $C(\theta) \cap W_1 \cap W_2 \cap \cdots \cap W_r$ for some walls W_i .*

Proof. Note that each cone C_T is the intersection of certain half spaces defined by walls. Consequently, each intersection of cones C_T for any finite set of trees T is an intersection of half spaces defined by walls. This shows that each facet of $C(\theta)$ is just the intersection of $C(\theta)$ with some half space of a wall. Since any face is the intersection of facets the result follows. $\square$

Note that the intersection of $C(\theta)$ with a wall W is *not* necessarily a face, see Section 6 and Example 2. Otherwise Theorem 5.1 were easier to prove. The next lemma will be useful to restrict to the case of generic weights. For simplicity we define $C(\theta) = \emptyset$ for θ not in C_Q .

Lemma 5.4. *Let θ be a weight in C_Q and let B be a sufficiently small neighbourhood of θ . In particular, it is sufficient that B intersects a wall W only if θ is in W . Then*

$$C(\theta) = \bigcap_{\xi \in B} C(\xi) = \bigcap_{i \in I} C(\theta^i),$$

where θ^i is a finite set of generic weights in B with $B \subseteq \bigcup C(\theta^i)$.

Proof. It is sufficient to show

$$\mathcal{T}(\theta) = \bigcup_{\xi \in B} \mathcal{T}(\xi) = \bigcup_{i \in I} \mathcal{T}(\theta^i).$$

To show the first equality let T be a tree admitting a regular flow ε with input ξ . Then each wall containing ξ also contains θ . Consequently, if the flow $\varepsilon^T(\xi)$ is regular, then $\varepsilon^T(\theta)$ must be also regular. Otherwise there were a wall separating ξ from θ , a contradiction to the assumption on B . The other inclusion is obvious, since θ is in B .

To show the second equality let T be a tree in $\mathcal{T}(\theta^i)$ for some generic weight θ^i . Then T is in $\mathcal{T}(\theta)$ by the same argument as above (θ^i is a particular weight in B). Let T be a tree in $\mathcal{T}(\theta)$. We have to show, there exists some generic weight θ^i with T in $\mathcal{T}(\theta^i)$. Consider a flow ε with $\varepsilon(\alpha) = a_\alpha > 0$ for each arrow α in T . Then the input of the flow $\varepsilon^T(\theta) + \lambda\varepsilon$ is generic for a generic choice of the numbers a_α . Further for a sufficiently small real number λ this flow is in B . Take θ^i to be the input of this flow for the particular choice of a_α and λ , then T is in $\mathcal{T}(\theta^i)$. Finally, the set I can be chosen finite, since there exist only finitely many different cones $C(\theta)$ by definition (the set of trees is a finite set). $\square$

Lemma 5.5. *Let θ be a generic weight. Let F be a face of $C(\theta)$ and let η be a weight in the relative interior of F . Then $C(\eta)$ coincides with the face F of $C(\theta)$.*

Proof. Let η be a weight and $B(\eta)$ be a sufficiently small neighbourhood of η . In particular, $B(\eta)$ intersects a wall W precisely if η is already in W and $B(\eta)$ is contained in C_Q . Then $C(\eta) = \bigcap_{\xi \in B(\eta)} C(\xi)$ (Lemma 5.4). On the other hand

$$C(\eta) = \bigcap_{\xi \in B(\eta)} C(\xi) = \bigcap_{j=1}^l C(\theta^j)$$

for a finite number of generic weights in $B(\eta)$. Then, we choose an order of the weights θ^i so that $\theta = \theta^1$, θ^2 is separated from θ^1 by precisely one wall W_1 . Next we choose θ^3 so that it is separated from θ^1 or θ^2 by precisely one wall W_2 . Finally, we obtain a sequence of weights and walls, so that θ^j is separated from some θ^i for $i < j$ by precisely one wall W_i . We apply Lemma 5.2 to this sequence. Thus

$$\bigcap_{j=1}^l C(\theta^j) = C(\theta) \cap \bigcap_{i=1}^l W_i.$$

Note that some walls W_i may appear more than once. By construction, all the walls W_i contain the face F and the result is a face of $C(\theta)$ which coincides with $C(\eta)$. $\square$

Lemma 5.6. *Let θ^i for i in I be a finite number of generic weights. Let η be weight in the relative interior of the intersection $\bigcap_{i \in I} C(\theta^i)$. Then $C(\eta) = \bigcap_{i \in I} C(\theta^i)$ is a face of each of the cones $C(\theta^i)$ for i in I .*

Proof. Let η be weight in the relative interior of the intersection $\bigcap_{i=1}^r C(\theta^i)$. Let B be a sufficiently small neighbourhood of η , so that η is in the relative interior of $\bigcap_{\xi \in B} C(\xi)$ (see Lemma 5.4). In B exists a finite number of generic weights η^j for j in J , so that $B \subset \bigcup_{j \in J} C(\eta^j)$ and

$$\bigcap_{i \in I} C(\theta^i) = \bigcap_{j \in J} C(\eta^j),$$

since the weights η^j can be chosen so that each cone $C(\theta^i)$ for i in I coincides with a cone $C(\eta^j)$ for some j in J . Similarly to the argument in the proof of the lemma above, we can choose an order on J , so that each cone $C(\eta^j)$ is separated by precisely one inner wall from one cone $C(\eta^l)$ with $l < j$. Consequently, the same argument shows that

$$\bigcap_{j \in J} C(\eta^j) = C(\eta^1) \cap \bigcap_{i=1}^r W_i = C(\eta)$$

for a finite number of inner walls separating $C(\eta^1)$. Thus, $C(\eta)$ is a face of $C(\eta^1)$ and, consequently, also a face of each $C(\theta^i)$ for i in I . $\square$

Lemma 5.7. *Let θ and η be two weights in C_Q . Then the intersection of the two cones $C(\theta)$ and $C(\eta)$ is a face of each.*

Proof. Similarly as in the proof of the previous lemma, we can choose small neighbourhoods of B of θ and B' of η with

$$C(\theta) = \bigcap_{\xi \in B} C(\xi) = \bigcap_{i \in I} C(\theta^i) \quad \text{and} \quad C(\eta) = \bigcap_{\xi' \in B'} C(\xi') = \bigcap_{i \in I'} C(\eta^i)$$

for a finite number of generic weights θ^i , $i \in I$ and η^i , $i \in I'$. Then we apply the previous lemma to show that $C(\theta) \cap C(\eta)$ is a face of each cone $C(\xi)$ for ξ generic in B or B' . Moreover, $C(\theta)$ is a face of each $C(\xi)$ for ξ generic in B and $C(\theta) \cap C(\eta)$ is the intersection of $C(\xi)$ with a finite number of supporting walls W_i for $i = 1, \dots, r$. Thus $C(\theta) \cap C(\eta) = C(\theta) \cap \bigcap_{i=1}^r W_i$. Since any supporting hyperplane of a cone supports also each of its faces we conclude $C(\theta) \cap C(\eta)$ is a face of $C(\theta)$. The same argument applies to $C(\eta)$. $\square$

Lemma 5.8. *Let F be a face of some cone $\overline{C}(\theta)$. Then F coincides with the cone $\overline{C}(\eta)$ for some weight η in the relative interior of F .*

Proof. There are two cases: either F is contained in the boundary of C_Q or it is not contained in the boundary of C_Q . In the latter case $\overline{C}(\theta)$ is also not in the boundary, so it coincides with $C(\theta)$ and it is the face of some cone $C(\theta')$ for some generic θ' . Consequently, F is a face of $C(\theta')$ and of the form $C(\eta)$ for some weight η in the relative interior of F .

Assume F is contained in the boundary of C_Q . Let $W_1, \dots, W_r$ be the set of outer walls containing F , where $W_1, \dots, W_l$ is the set of outer walls containing $\overline{C}(\theta)$. Then $C(\eta)$ is a face of $C(\theta)$ since

$$C(\eta) = \bigcap C(\eta^j) \quad \text{and} \quad C(\theta) = \bigcap C(\theta^i)$$

for some generic weights η^j and θ^i (see Lemma 5.4). Then $C(\eta)$ is a face of $C(\theta)$. Further, all outer walls are supporting for any cone, in particular for $C(\eta)$ and $C(\theta)$. So $\overline{C}(\eta)$ is a face of $\overline{C}(\theta)$. $\square$

Lemma 5.9. *The intersection of two cones $\overline{C}(\theta)$ and $\overline{C}(\theta')$ is a face of each.*

Proof. First we assume θ and θ' are not in the boundary of C_Q . Then the proof follows by Lemma 5.7. If at least one is in the boundary, then we write $\overline{C}(\theta)$ as the intersection of $C(\theta)$ with some outer walls W_i and $\overline{C}(\theta')$ as the intersection of $C(\theta')$ with some outer walls V_i . The intersection of $C(\theta)$ with $C(\theta')$ is a face of both. Since any outer wall is a supporting wall for any of the cones $C(\theta)$, $C(\theta')$, $\overline{C}(\theta)$, and $\overline{C}(\theta')$ we proved the claim. $\square$

Proof of Theorem 5.1. First we note that the set of cones $\{\overline{C}(\theta)\}$ is a finite set, since we can, using the finite set of walls and the cones C_T , form only a finite number of possible intersections.

We show that each cone $\overline{C}(\theta)$ is strongly convex, rational and polyhedral. By Lemma 4.1 any cone C_T is strongly convex, so also any intersection. Moreover, any C_T is obviously polyhedral and any finite intersection is polyhedral as well, so $\overline{C}(\theta)$ is polyhedral. To show rationality we note that any intersection of a finite number of hyperplanes defined by rational equations admits a basis of solutions which is rational. Consequently also the cones $\overline{C}(\theta)$ are rational.

To finish the proof we note that property (1) of a fan is Lemma 5.8 and property (2) is Lemma 5.9. $\square$

Remark. Finally we note, that there exists a distinguished weight θ^c , the so-called *canonical weight*, which is the input of the canonical flow ε^c defined by $\varepsilon_\alpha^c = 1$ for all arrows α in Q_1 . So

$$\theta^c(q) = \sharp\{\alpha \mid s(\alpha) = q\} - \sharp\{\alpha \mid t(\alpha) = q\}.$$

The advantage of the canonical flow is, that the polytope $\Delta(\theta^c)$ is reflexive in the sense of Batyrev [2] (see [1] for a proof and further properties). The canonical weight is an element of any (t, t) -wall and it lies in the positive half space of any (t^+, t^-) -wall with $t^+ > t^-$. Thus it is in a certain sense the central weight in C_Q and the cone $\overline{C}(\theta^c)$ is the central cone in C_Q (in Fig. 11 it is the polytope on the right-hand side in the middle line).

6. Examples

Example 1. An example of a polytope defined by a quiver with a large symmetry group is the following one. It is distinguished in the class of three-dimensional quiver polytopes by two properties: it is reflexive and it has the maximal number of facets a quiver polytope can have. Note that the number of facets is 12 and the number of vertices is 14. In Fig. 4 we show the quiver, the polytope Δ , the canonical weight and the isomorphism classes of trees with their flows. The possible embeddings of these trees correspond to the vertices of the polytope Δ . So there are two different

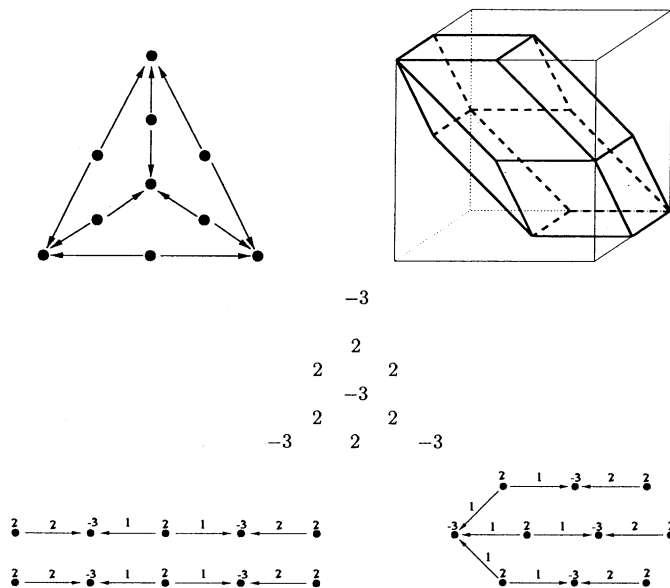


Fig. 4.

kinds of vertices: those which lie in the intersection of four facets (corresponding to the first type of trees, we obtain six such vertices) and those which lie on the intersection of three facets (corresponding to the second type of trees, we obtain eight such vertices).

For further investigation of this example we introduce the following notation. We denote the four sinks of the quiver by $1, \dots, 4$ and we denote the six sources of the quiver by tuples $(1, 2), (1, 3), \dots, (3, 4)$, according to their connecting arrows to the four sinks. These tuples we consider up to their order. Thus we identify such a tuple (i, j) with its underlying set $\{i, j\}$. In a similar way, we denote the arrows by $\alpha_{i,j}$, where $s(\alpha_{i,j}) = \{i, j\}$ and $t(\alpha_{i,j}) = i$. Thus, arrows are indexed by ordered tuples of different numbers i, j for $1 \leq i, j \leq 4$. The symmetric group S_4 of permutations of $\{1, \dots, 4\}$ acts on the quiver and also on the polytope. It acts transitively on the arrows, on the sinks and on the sources. We wish to determine the stabilisers and a fundamental region for the action of S_4 on Δ .

Let (i, j) be an ordered tuple, $\{i, j\}$ an unordered tuple and i be an element. Then the stabiliser of these elements are

$$\text{Stab}((i, j)) = \{\sigma \in S_4 \mid \sigma(i) = i, \sigma(j) = j\} = S(\{1, \dots, 4\} \setminus \{i, j\}) \simeq S_2,$$

$$\text{Stab}(\{i, j\}) = \{\sigma \in S_4 \mid \sigma(\{i, j\}) \subset \{i, j\}\} = S(\{i, j\}) \times S(\{1, \dots, 4\} \setminus \{i, j\}),$$

$$\text{Stab}(\{i\}) = \{\sigma \in S_4 \mid \sigma(i) = i\} = S(\{1, \dots, 4\} \setminus \{i\}) \simeq S_3,$$

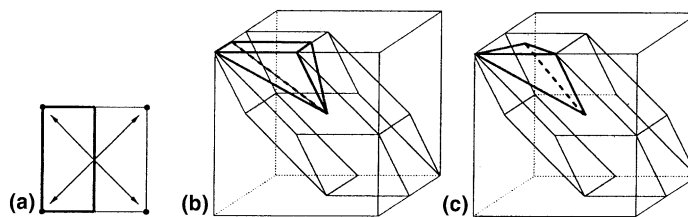


Fig. 5. (a) The action of the stabiliser on a facet of Δ . (b) Fundamental region for the S_4 action. (c) A fundamental region for the $\text{Aut}(\Delta)$ -action.

respectively. We see that on any facet there is a nontrivial stabiliser isomorphic to S_2 , so a fundamental region is just a cone over a half-facet intersected with Δ . We take the the following set as a fundamental region: let $F_{1,2}$ be the facet defined by $\alpha_{1,2}$. This facet contains four vertices (see Fig. 5). We choose a fundamental region for the action of the stabiliser on this facet. The stabiliser is isomorphic to S_2 and it acts via the central symmetry as shown in the figure. Thus the half-facet is a fundamental region for the action of S_2 on this facet and, consequently, the cone over this facet is a fundamental region for the action of S_{24} in the polytope Δ .

Note that the polytope Δ admits a larger symmetry group. There is an additional symmetry defined by central reflection. So the group of lattice automorphisms $\text{Aut}(\Delta)$ of Δ is a semidirect product of S_4 with $\mathbb{Z}_2$. A fundamental region for this action can be found on the right-hand side of Fig. 5.

Example 2. We consider the polytope of the complete quiver with four vertices K_4 (see Fig. 6) whose canonical weight is given by $(3, 1, -1, -3)$. To describe this polytope we use the map f to $\mathbb{R}$ defined by sending a flow ε to its component $x = \varepsilon(\alpha_{(1,4)})$. Obviously $0 \leq x \leq 3$ and conversely for any such x the polytope $f^{-1}(x)$

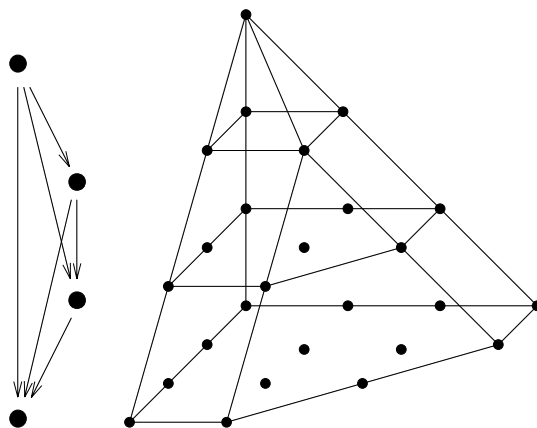


Fig. 6.

is nonempty. Note that $[0, 3]$ can be stratified into a finite number of subintervals, so that the inverse $f^{-1}(x)$ is combinatorially equivalent for any x in the inner part of this stratum. In this example we get a decomposition into two strata: $[0, 2]$ and $[2, 3]$. Using this description, we obtain: $f^{-1}(3)$ is a point, $f^{-1}(2)$ is the rectangle $[0, 1] \times [0, 1]$ and $f^{-1}(0)$ is the convex hull of $(0, 0)$, $(3, 0)$, $(3, 1)$, $(1, 3)$, and $(0, 3)$. If we use the x -coordinate for the height of the polytope we obtain the polytope in Fig. 6.

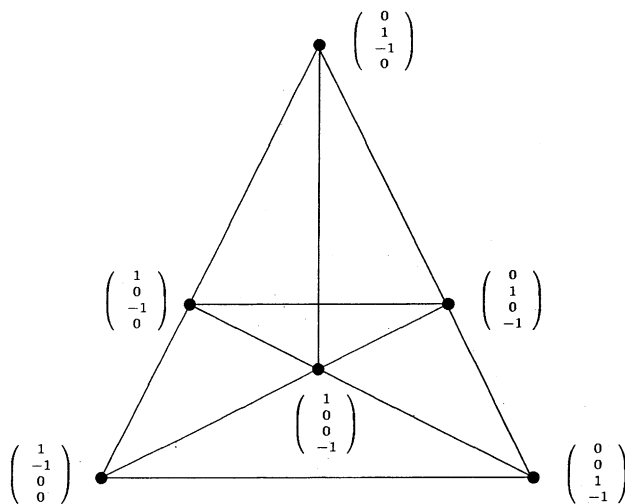


Fig. 7. A base of the C_Q with its elementary weights.

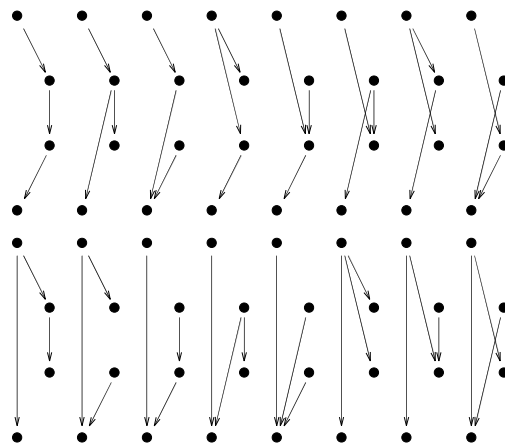


Fig. 8. The tress of the quiver K_4 .

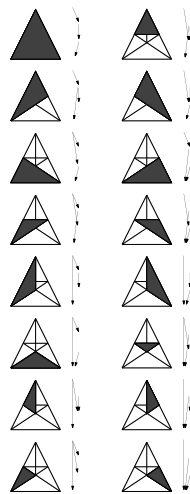


Fig. 9. The tree T and the cone C_T .

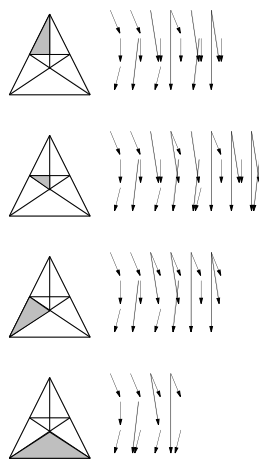


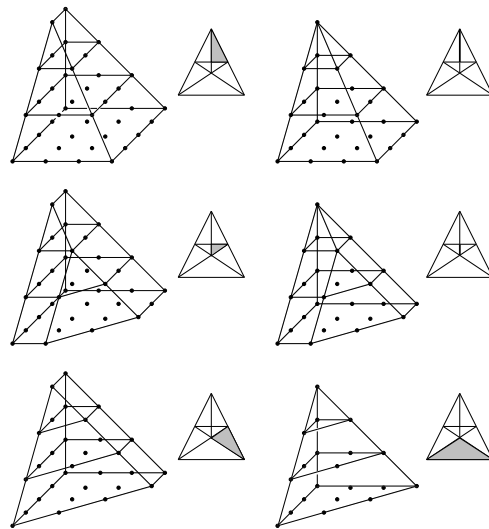
Fig. 10. The cone $C(\theta)$ and the set $T(\theta)$.

Remark on the Figs. 7–11. All the Figs. 7–11 are related to Example 2, so we always consider the quiver K_4 .

In Fig. 7 we show the intersection of the cone C_Q with a sufficiently generic hyperplane, so the figure shows a base of the cone, together with the elementary weights. Here one can already see the cones $C(\theta)$ for the various weights θ .

In Fig. 8 we list all trees of K_4 , there are precisely 16 different trees.

In Fig. 9 we indicate the cones C_T for the 16 trees shown in Fig. 8.

Fig. 11. Certain polytopes $\Delta(\theta)$.

In Fig. 10 we determine the sets $\mathcal{T}(\theta)$ for the weights θ in the interior of the indicated region.

Finally in Fig. 11 we show a polytope $\Delta(\theta)$ for some weight in the indicated region or on some boundary of some cone, respectively.

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